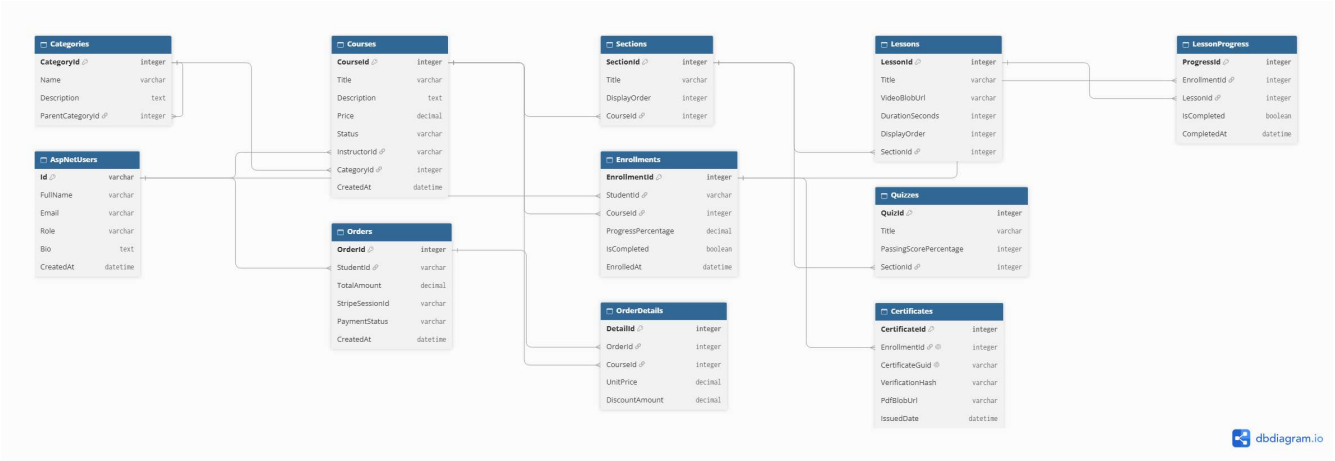
Database Design Specifications & Schemas

1. Entity-Relationship Diagram (ERD)



2. Logical Database Schema

Table Details & Data Types

1. **`AspNetUsers`** (Identity Table)

    `Id` (`nvarchar(450)`, Primary Key, NOT NULL)

    `FullName` (`nvarchar(150)`, NOT NULL)

    `Email` (`nvarchar(256)`, NOT NULL)

    `Role` (`nvarchar(50)`, NOT NULL)

    `Bio` (`nvarchar(MAX)`, NULL)

    `CreatedAt` (`datetime2`, NOT NULL, Default: `GETUTCDATE()`)

    -

2. **Categories**

   `CategoryId` (`int`, Primary Key, Identity, NOT NULL)

   `Name` (`nvarchar(100)`, NOT NULL)

   `Description` (`nvarchar(MAX)`, NULL)

   `ParentCategoryId` (`int`, Foreign Key -> `Categories.CategoryId`, NULL)

3. **Courses**

   `CourseId` (`int`, Primary Key, Identity, NOT NULL)

   `Title` (`nvarchar(200)`, NOT NULL)

   `Description` (`nvarchar(MAX)`, NULL)

   `Price` (`decimal(18,2)`, NOT NULL)

   `Status` (`nvarchar(50)`, NOT NULL)

   `InstructorId` (`nvarchar(450)`, Foreign Key -> `AspNetUsers.Id`, NOT NULL)

   `CategoryId` (`int`, Foreign Key -> `Categories.CategoryId`, NOT NULL)

   `CreatedAt` (`datetime2`, NOT NULL, Default: `GETUTCDATE()`)

4. **Sections**

   `SectionId` (`int`, Primary Key, Identity, NOT NULL)

   `Title` (`nvarchar(200)`, NOT NULL)

   `DisplayOrder` (`int`, NOT NULL)

   `CourseId` (`int`, Foreign Key -> `Courses.CourseId`, NOT NULL)

5. **Lessons**

   `LessonId` (`int`, Primary Key, Identity, NOT NULL)

   `Title` (`nvarchar(200)`, NOT NULL)

   `VideoBlobUrl` (`nvarchar(500)`, NOT NULL)

   `DurationSeconds` (`int`, NOT NULL)

`DisplayOrder` (`int`, NOT NULL)

`SectionId` (`int`, Foreign Key -> `Sections.SectionId`, NOT NULL)

6. **`Quizzes`**

   `QuizId` (`int`, Primary Key, Identity, NOT NULL)

   `Title` (`nvarchar(200)`, NOT NULL)

   `PassingScorePercentage` (`int`, NOT NULL)

   `SectionId` (`int`, Foreign Key -> `Sections.SectionId`, NOT NULL)

7. **`Enrollments`**

   `EnrollmentId` (`int`, Primary Key, Identity, NOT NULL)

   `StudentId` (`nvarchar(450)`, Foreign Key -> `AspNetUsers.Id`, NOT NULL)

   `CourseId` (`int`, Foreign Key -> `Courses.CourseId`, NOT NULL)

   `ProgressPercentage` (`decimal(5,2)`, NOT NULL, Default: `0.00`)

   `IsCompleted` (`bit`, NOT NULL, Default: `0`)

   `EnrolledAt` (`datetime2`, NOT NULL, Default: `GETUTCDATE()`)

8. **`LessonProgress`**

   `ProgressId` (`int`, Primary Key, Identity, NOT NULL)

   `EnrollmentId` (`int`, Foreign Key -> `Enrollments.EnrollmentId`, NOT NULL)

   `LessonId` (`int`, Foreign Key -> `Lessons.LessonId`, NOT NULL)

   `IsCompleted` (`bit`, NOT NULL, Default: `0`)

   `CompletedAt` (`datetime2`, NULL)

9. **`Orders`**

   `OrderId` (`int`, Primary Key, Identity, NOT NULL)

   `StudentId` (`nvarchar(450)`, Foreign Key -> `AspNetUsers.Id`, NOT NULL)

   `TotalAmount` (`decimal(18,2)`, NOT NULL)

`StripeSessionId` (`nvarchar(200)`, NOT NULL)

`PaymentStatus` (`nvarchar(50)`, NOT NULL)

`CreatedAt` (`datetime2`, NOT NULL, Default: `GETUTCDATE()`)

10. **`OrderDetails`**

    `DetailId` (`int`, Primary Key, Identity, NOT NULL)

    `OrderId` (`int`, Foreign Key -> `Orders.OrderId`, NOT NULL)

    `CourseId` (`int`, Foreign Key -> `Courses.CourseId`, NOT NULL)

    `UnitPrice` (`decimal(18,2)`, NOT NULL)

    `DiscountAmount` (`decimal(18,2)`, NOT NULL, Default: `0.00`)

11. **`Certificates`**

    `CertificateId` (`int`, Primary Key, Identity, NOT NULL)

    `EnrollmentId` (`int`, Foreign Key -> `Enrollments.EnrollmentId`, Unique, NOT NULL)

    `CertificateGuid` (`uniqueidentifier`, Unique, NOT NULL)

    `VerificationHash` (`nvarchar(128)`, NOT NULL)

    `PdfBlobUrl` (`nvarchar(500)`, NOT NULL)

    `IssuedDate` (`datetime2`, NOT NULL, Default: `GETUTCDATE()`)

3. Physical Database Schema (SQL Server Script)

```
-- Create Database
CREATE DATABASE EduHubDb;
GO
USE EduHubDb;
GO

-- 1. AspNetUsers (Identity Table)
CREATE TABLE AspNetUsers (
    Id NVARCHAR(450) PRIMARY KEY,
    FullName NVARCHAR(150) NOT NULL,
    Email NVARCHAR(256) NOT NULL,
    Role NVARCHAR(50) NOT NULL,
    Bio NVARCHAR(MAX) NULL,
    CreatedAt DATETIME2 NOT NULL DEFAULT GETUTCDATE()
```

```
);
GO

-- 2. Categories
CREATE TABLE Categories (
    CategoryId INT IDENTITY(1,1) PRIMARY KEY,
    Name NVARCHAR(100) NOT NULL,
    Description NVARCHAR(MAX) NULL,
    ParentCategoryId INT NULL,
    CONSTRAINT FK_Categories_Parent FOREIGN KEY (ParentCategoryId)
REFERENCES Categories(CategoryId) ON DELETE NO ACTION
);
GO

-- 3. Courses
CREATE TABLE Courses (
    CourseId INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(200) NOT NULL,
    Description NVARCHAR(MAX) NULL,
    Price DECIMAL(18,2) NOT NULL,
    Status NVARCHAR(50) NOT NULL,
    InstructorId NVARCHAR(450) NOT NULL,
    CategoryId INT NOT NULL,
    CreatedAt DATETIME2 NOT NULL DEFAULT GETUTCDATE(),
    CONSTRAINT FK_Courses_Instructor FOREIGN KEY (InstructorId)
REFERENCES AspNetUsers(Id) ON DELETE CASCADE,
    CONSTRAINT FK_Courses_Category FOREIGN KEY (CategoryId)
REFERENCES Categories(CategoryId) ON DELETE CASCADE
);
GO

-- 4. Sections
CREATE TABLE Sections (
    SectionId INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(200) NOT NULL,
    DisplayOrder INT NOT NULL,
    CourseId INT NOT NULL,
    CONSTRAINT FK_Sections_Course FOREIGN KEY (CourseId) REFERENCES
Courses(CourseId) ON DELETE CASCADE
);
GO

-- 5. Lessons
CREATE TABLE Lessons (
    LessonId INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(200) NOT NULL,
    VideoBlobUrl NVARCHAR(500) NOT NULL,
    DurationSeconds INT NOT NULL,
    DisplayOrder INT NOT NULL,
    SectionId INT NOT NULL,
    CONSTRAINT FK_Lessons_Section FOREIGN KEY (SectionId) REFERENCES
Sections(SectionId) ON DELETE CASCADE
);
GO

-- 6. Quizzes
CREATE TABLE Quizzes (
    QuizId INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(200) NOT NULL,
    PassingScorePercentage INT NOT NULL,
    SectionId INT NOT NULL,
```

```
    CONSTRAINT FK_Quizzes_Section FOREIGN KEY (SectionId) REFERENCES
Sections(SectionId) ON DELETE CASCADE
);
GO

-- 7. Enrollments
CREATE TABLE Enrollments (
    EnrollmentId INT IDENTITY(1,1) PRIMARY KEY,
    StudentId NVARCHAR(450) NOT NULL,
    CourseId INT NOT NULL,
    ProgressPercentage DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    IsCompleted BIT NOT NULL DEFAULT 0,
    EnrolledAt DATETIME2 NOT NULL DEFAULT GETUTCDATE(),
    CONSTRAINT FK_Enrollments_Student FOREIGN KEY (StudentId)
REFERENCES AspNetUsers(Id) ON DELETE NO ACTION,
    CONSTRAINT FK_Enrollments_Course FOREIGN KEY (CourseId)
REFERENCES Courses(CourseId) ON DELETE CASCADE
);
GO

-- 8. LessonProgress
CREATE TABLE LessonProgress (
    ProgressId INT IDENTITY(1,1) PRIMARY KEY,
    EnrollmentId INT NOT NULL,
    LessonId INT NOT NULL,
    IsCompleted BIT NOT NULL DEFAULT 0,
    CompletedAt DATETIME2 NULL,
    CONSTRAINT FK_LessonProgress_Enrollment FOREIGN KEY (EnrollmentId)
REFERENCES Enrollments(EnrollmentId) ON DELETE CASCADE,
    CONSTRAINT FK_LessonProgress_Lesson FOREIGN KEY (LessonId)
REFERENCES Lessons(LessonId) ON DELETE NO ACTION
);
GO

-- 9. Orders
CREATE TABLE Orders (
    OrderId INT IDENTITY(1,1) PRIMARY KEY,
    StudentId NVARCHAR(450) NOT NULL,
    TotalAmount DECIMAL(18,2) NOT NULL,
    StripeSessionId NVARCHAR(200) NOT NULL,
    PaymentStatus NVARCHAR(50) NOT NULL,
    CreatedAt DATETIME2 NOT NULL DEFAULT GETUTCDATE(),
    CONSTRAINT FK_Orders_Student FOREIGN KEY (StudentId) REFERENCES
AspNetUsers(Id) ON DELETE NO ACTION
);
GO

-- 10. OrderDetails
CREATE TABLE OrderDetails (
    DetailId INT IDENTITY(1,1) PRIMARY KEY,
    OrderId INT NOT NULL,
    CourseId INT NOT NULL,
    UnitPrice DECIMAL(18,2) NOT NULL,
    DiscountAmount DECIMAL(18,2) NOT NULL DEFAULT 0.00,
    CONSTRAINT FK_OrderDetails_Order FOREIGN KEY (OrderId) REFERENCES
Orders(OrderId) ON DELETE CASCADE,
    CONSTRAINT FK_OrderDetails_Course FOREIGN KEY (CourseId)
REFERENCES Courses(CourseId) ON DELETE NO ACTION
);
GO
```

```
-- 11. Certificates
CREATE TABLE Certificates (
    CertificateId INT IDENTITY(1,1) PRIMARY KEY,
    EnrollmentId INT NOT NULL UNIQUE,
    CertificateGuid UNIQUEIDENTIFIER NOT NULL UNIQUE,
    VerificationHash NVARCHAR(128) NOT NULL,
    PdfBlobUrl NVARCHAR(500) NOT NULL,
    IssuedDate DATETIME2 NOT NULL DEFAULT GETUTCDATE(),
    CONSTRAINT FK_Certificates_Enrollment FOREIGN KEY (EnrollmentId)
REFERENCES Enrollments(EnrollmentId) ON DELETE CASCADE
);
GO

-- Performance Indexes
CREATE INDEX IX_Courses_CategoryId ON Courses(CategoryId);
CREATE INDEX IX_Courses_InstructorId ON Courses(InstructorId);
CREATE INDEX IX_Lessons_SectionId ON Lessons(SectionId);
CREATE INDEX IX_Enrollments_StudentId ON Enrollments(StudentId);
GO
```